# Sungmin Yun

Computer Architectures and Systems for AI

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## About Me

I am a **Senior Deep Learning Performance Architect at NVIDIA**, focusing on AI acceleration, performance analysis, and optimization. I have extensive hands-on experience in **accelerator design, hardware profiling, performance modeling, and kernel programming**, with strong expertise in **modern AI architectures** (e.g., MoE, MLA, RAG, diffusion). My research has focused on **maximizing hardware utilization** for LLM serving and accelerating **Multi-LoRA inference** and **RAG pipelines**. I also developed an **open-source end-to-end LLM performance simulator**, LLMSimulator, enabling rapid evaluation of emerging hardware and LLM architectures. My work aims to **maximize the performance and efficiency of AI systems** through hardware–software co-design.

## Selected Publications

**[MICRO 2024] Duplex: A Device for Large Language Models with Mixture of Experts, Grouped Query Attention, and Continuous Batching**, **S. Yun**, K. Kyung, J. Cho, J. Choi, J. Kim, B. Kim, S. Lee, K. Sohn, J. H. Ahn

**[ICS 2024] CLAY: CXL-based Scalable NDP Architecture Accelerating Embedding Layers**, **S. Yun**, H. Nam, K. Kyung, J. Park, B. Kim, Y. Kwon, E. Lee, J. H. Ahn

**[ICS 2026] Clone: A Collaborative Multi-device System for Retrieval-Augmented Generation over CXL**, S. Ko, W. Doh, E. Na, H. Shim, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, E. Lee, J. H. Ahn (Co-corresponding author)

## Experience

**NVIDIA** | **Senior Deep Learning Performance Architect** Sep 2026 – Present

- **Deep learning performance analysis.** Analyze the performance of deep learning applications, focusing on AI acceleration and performance optimization.

**Seoul National University** | **Postdoctoral Researcher** Sep 2025 – Sep 2026

- **LLM serving systems.** Developed system-level guidelines and optimizations for LLM serving, including Multi-head Latent Attention (MLA) and Mixture-of-Experts (MoE) architectures.
- **Multi-LoRA inference at scale.** Led the end-to-end design and implementation of an accelerator for Multi-LoRA inference.
- **RAG pipeline acceleration.** Co-designed search algorithms and hardware for large-scale retrieval-augmented generation pipelines.

**Samsung Electronics** | **Intern** Mar 2023 – Aug 2023

- **HBM-PIM controller design.** Designed and implemented a hardware controller in RTL for general matrix–vector multiplication (GEMV) using HBM-PIM.
- **LLM accelerator architecture.** Analyzed the architectural limitations of HBM-PIM and proposed Duplex, an LLM accelerator integrating custom HBMs with xPUs. This work was later published at MICRO 2024.

## Education

**Seoul National University**, Ph.D. in Artificial Intelligence GPA: 4.1/4.3 Aug 2020 – Aug 2025

**Yonsei University**, B.S. in Integrated Information Technology GPA: 3.6/4.3 Mar 2017 – Feb 2020

## Technical Skills

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| **Programming languages** | C/C++, CUDA, Python, System Verilog |
| **ML/LLM Frameworks** | Pytorch, vLLM, SGLang, TensorRT-LLM |
| **Profiling, Analysis, Simulators** | Nsight Compute, Nsight Systems, VTune, Ramulator |

## Honors & Awards

**Best Ph.D. Dissertation Award** Aug 2025
Seoul National University

**Best of CAL Award** Feb 2023
29th IEEE International Symposium on High-Performance Computer Architecture

## Invited Presentations

Poster presentation at Samsung AI Forum Nov 2023
Oral presentation on Best of CAL session at IEEE International Symposium on High-Performance Computer Architecture Feb 2023

## Program Committee and Paper Review

Lightweight Program Committee Member, ISCA 2026
IEEE Transactions on Computers
IEEE Transactions on Circuits and Systems for Artificial Intelligence
IEEE Computer Architecture Letters
Future Generation Computer Systems

## Additional Publications

**[IEEE CAL 2025] SSD Offloading for LLM Mixture-of-Experts Weights Considered Harmful in Energy Efficiency**, K. Kyung, **S. Yun**, J. H. Ahn

**[IEEE CAL 2025] COSMOS: A CXL-Based Full In-Memory System for Approximate Nearest Neighbor Search**, S. Ko, H. Shim, W. Doh, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, J. H. Ahn

**[HPCA 2025] Anaheim: Architecture and Algorithms for Processing Fully Homomorphic Encryption in Memory**, J. Kim, **S. Yun**, H. Ji, W. Choi, S. Kim, J. H. Ahn

**[IEEE TC 2023] GraNDe: Efficient Near-Data Processing Architecture for Graph Neural Networks**, **S. Yun**, H. Nam, J. Park, B. Kim, J. H. Ahn, E. Lee

**[IEEE CAL 2022] GraNDe: Near-Data Processing Architecture With Adaptive Matrix Mapping for Graph Convolutional Networks**, **S. Yun**, B. Kim, J. Park, H. Nam, J. H. Ahn, E. Lee
Best of CAL Award

**[MICRO 2021] TRiM: Enhancing Processor-Memory Interfaces with Scalable Tensor Reduction in Memory**, J. Park, B. Kim, **S. Yun**, E. Lee, M. Rhu, J. H. Ahn

## Patents

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| • Heterogeneous Computing Memory Device and Artificial Intelligence Computing System Including the Same | US<br>KR (10-2025-0050877) |
| • Memory Device Capable of Dynamically Switching Between Error Correction Code and Error Detection Code in On-Die Error Correction Code | KR Granted (10-2797560)<br>PCT (PCT/KR2021/019871) |

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| • Memory Module and Computing System Including the Same | US (19/017,235)<br>KR (10-2024-0060980)<br>CN (202510116255.2) |
| • Memory Controller and Operation Method Thereof | US (19/002,383)<br>KR (10-2024-0050775)<br>CN (202411903289.8) |
| • Method and Apparatus for Memory Device Including Processing-in-Memory (PIM) | US (19/446,242)<br>KR (10-2025-0026131) |
| • Memory System and Operating Method Thereof | KR (10-2025-0195434) |